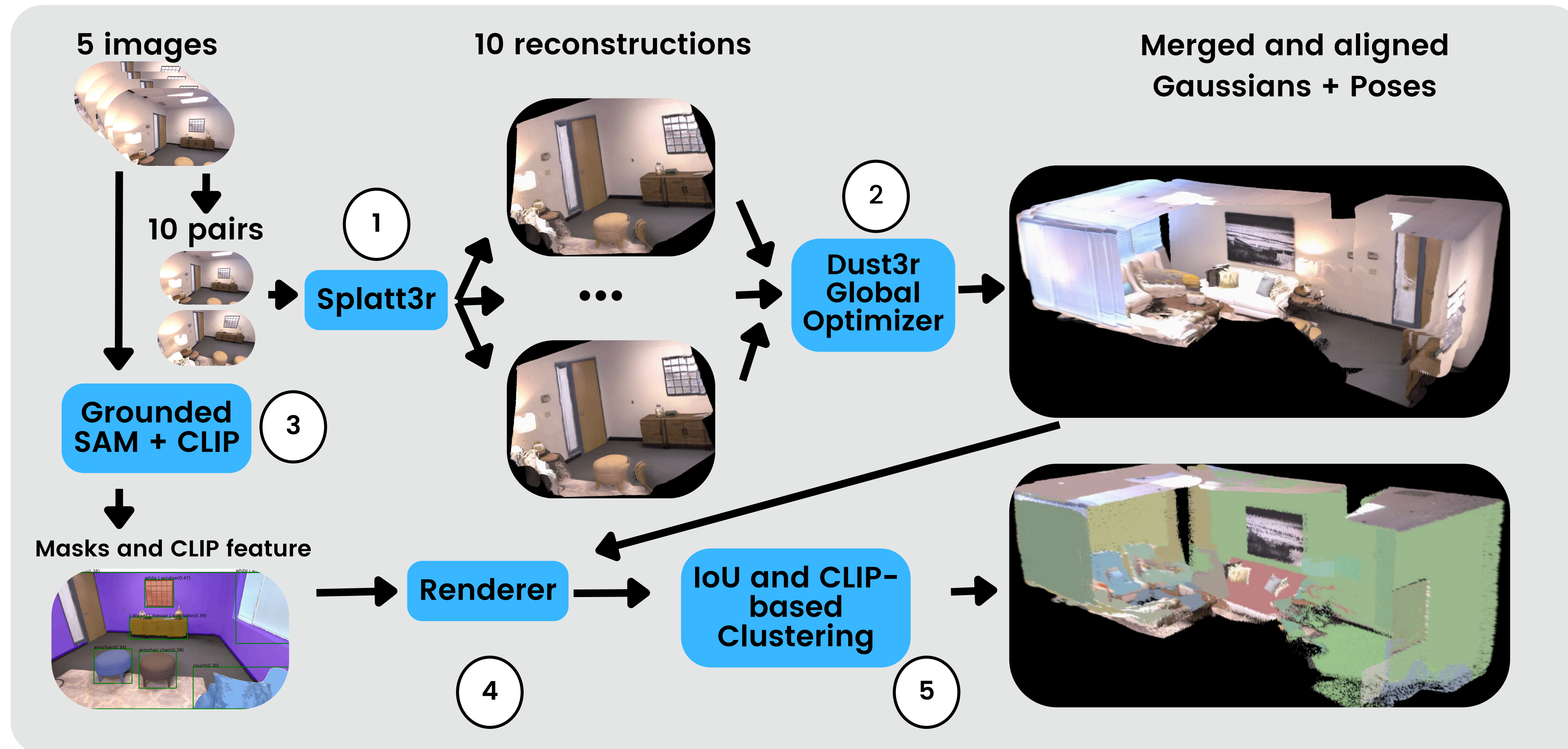


Extending Splatt3R: Multi-View Semantic Gaussian Splatting

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Pipeline Overview



Our Contributions:

- Predict dense 3D Gaussians from 10 image pairs using Splatt3r.
- Align all reconstructions and poses via DUST3R global optimization and ICP.
- Associate masks and CLIP labels with Gaussians using projection.
- Cluster Gaussians across views using IoU and CLIP similarity.

Score-based Semantic Clustering

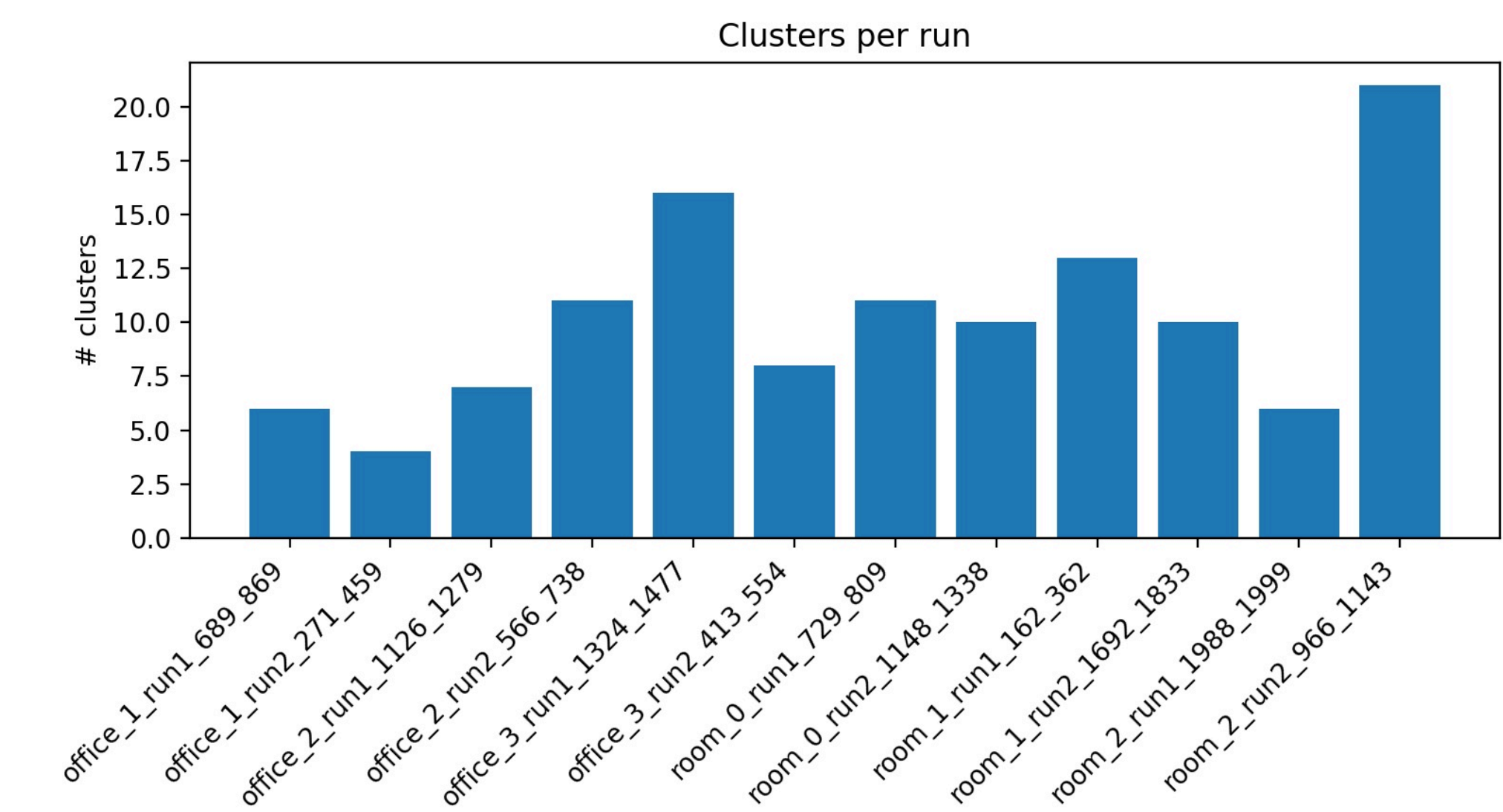
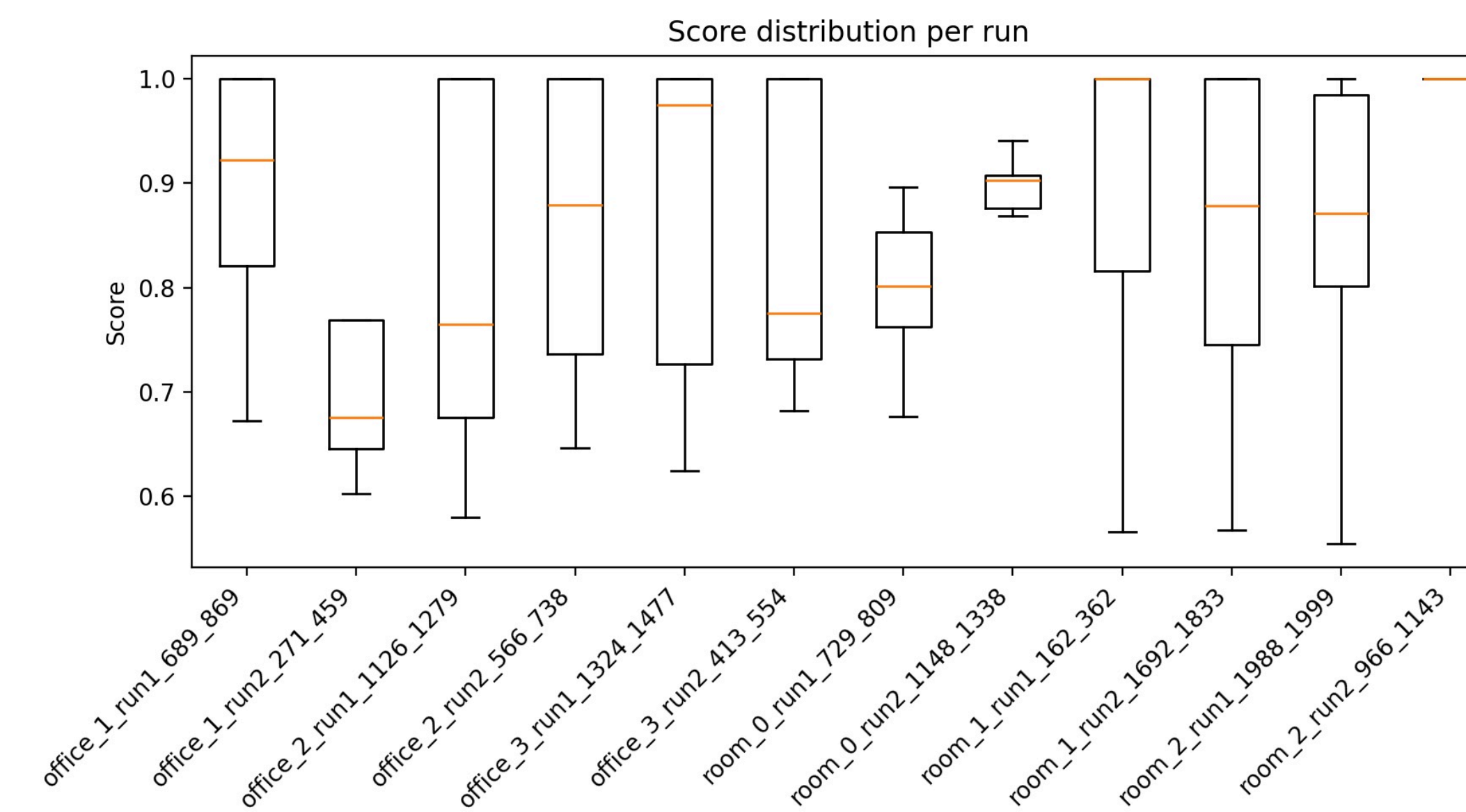
Merge cross-view Gaussian clusters based on score:

$$\text{Score} = \frac{1}{2} (\text{IoU}(\mathbf{a}, \mathbf{b}) + \text{CLIPSim}(\mathbf{f}_a, \mathbf{f}_b))$$



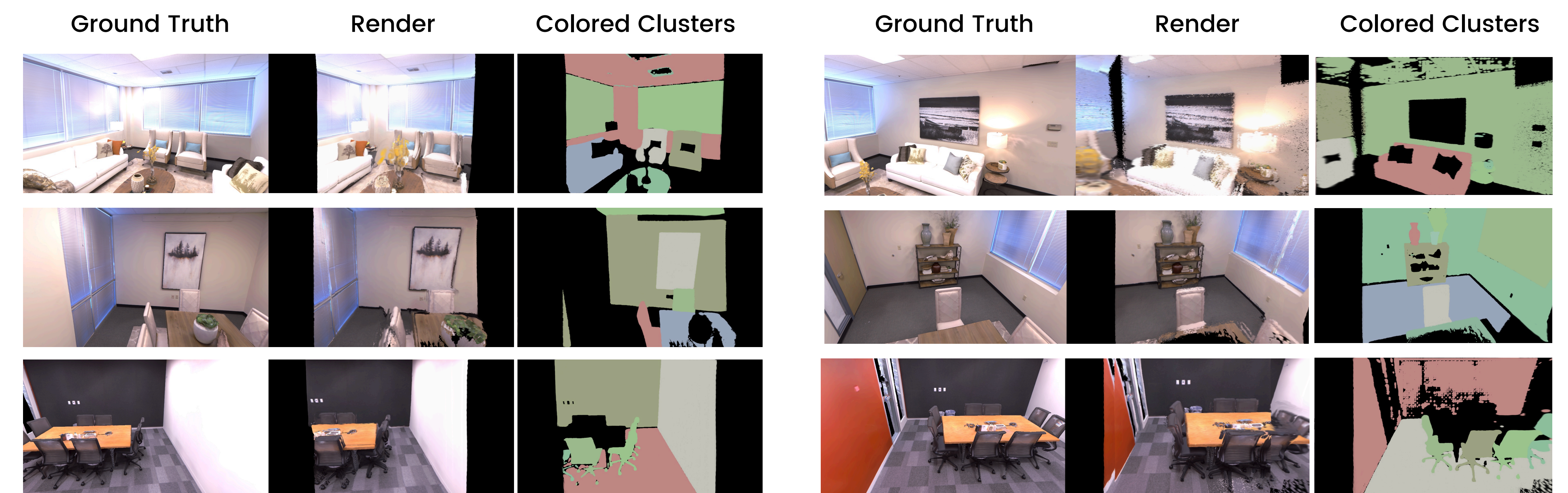
Results

Clustering Confidence



Qualitative Results

Render Comparison



Prompted Novel View of Clusters

